CMR ENGINEERING COLLEGE

UGC AUTONOMOUS

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II-B.TECH II-SEM (R22) I-MID Examinations-March-2026 **Date: 16/02/2026**

Subject: Computer Oriented Statistical Methods **Time: 10:00 TO 12:00 PM**

Branch: CSE **Marks: 30 M**

Answer all Questions in Part-A & Answer any FOUR Questions in Part-B

PART-A **5x2=10M**

1. State Baye's Theorem. (BL - 1) (CO 1)
2. The probability density function of a continuous random variable X is given by $f(x) = ce^{-|x|}, -\infty < x < \infty$. Show that $c = \frac{1}{2}$. (BL - 2) (CO 1)
3. If the mean of Binomial distribution is 3 and variance 9/4, obtain the value of n. (BL - 1) (CO 2)
4. If the mean of a Poisson distribution is 3, then find P (X = 0). (BL - 1) (CO 2)
5. Write Confidence Interval and Maximum Error formulae for large samples. (BL - 1) (CO 3)

PART - B **4x5 =20 M**

6. In a bolt factory machines A, B, C manufacture 20%, 30%, and 50% of the total of their output and 6%,3% and 2% are defective. A bolt is drawn at random and found to be defective. Find the probabilities that it is manufactured from
   i) Machine A ii) Machine B iii) Machine C (BL - 5) (CO 1)

7. A random variable X has the following probability function:

x:	0	1	2	3	4	5	6	7
p(x):	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

   Find the value of (a) k (b) Mean (c) variance (BL - 5) (CO 1)

8. Four coins are tossed 160 times. The number of times x heads occur is given below.

x	0	1	2	3	4
No. of times	8	34	69	43	6

   Fit a Binomial distribution to this data on the hypothesis that coins are unbiased. (BL - 4) (CO 2)

9. In a sample of 1000cases, the mean of a certain test is 14 and standard deviation is 2.5. Assuming the distribution to be normal, find
   i) How many students score between 12 and 15
   ii) How many students score above 18
   iii) How many students score below 18. (BL - 1) (CO 2)

10. Samples of size 3 are taken from the population 3,6,9,15,27 without replacement. Find
    a. The mean of the population.
    b. The standard deviation of the population.
    c. The mean of the sampling distribution of means and
    d. The standard deviation of the sampling distribution of means. (BL - 1) (CO 2)

11. The mean and standard deviation of the population are 11795 and 14054. If n=50, find 95% confidence interval for mean.